Sentiment analysis: Is a text positive or negative? This is an out-of-box API provided in transformer libraries.

```
from transformers import pipeline
sentiment = pipeline('sentiment-analysis')
result = sentiment('The food was good. Will have to try
again')
print(result)
result = sentiment('The shirt I bought is nice, but it seems
to be expensive')
print(result)

The Output is:
[{'label': 'POSITIVE', 'score': 0.9995126724243164}]
[{'label': 'NEGATIVE', 'score': 0.9585908651351929}
```

Named entity recognition (NER): This deals with extracting entities from a given sentence.

```
ner = pipeline('ner')
result = ner('Kolkata is known to be the one of the great
city in India.')
print(result)

The output is:
[{'word': 'Kolkata', 'score': 0.9992743730545044, 'entity':
'I-LOC', 'index': 1, 'start': 0, 'end': 7}, {'word': 'India',
'score': 0.992446780204773, 'entity': 'I-LOC', 'index': 13,
'start': 51, 'end': 56}]
```

Text summarisation: Summarisation takes in a passage as input and tries to summarise it.

```
summarizer = pipeline("summarization")
my_text="""On this day in 2011, India won the World Cup on
home soil to send the entire country into euphoria.
India had previously lifted the prized trophy in 1983 and had
to wait a long time before doing it again.
Led by MS Dhoni, India registered their second ODI title
triumph by beating Sri Lanka in the final at the Wankhede
Stadium
in Mumbai on April 2, 2011. That day will be remembered for
many a magical moments -- Gautam Gambhir's brilliant knock,
Dhoni's iconic six, Sachin Tendulkar being carried around the
ground by his teammates, to name a few."""
summarizer = pipeline("summarization")
summarized_10 = summarizer(my_text, min_length=5, max_
length=10)
summarized_20= summarizer(my_text, min_length=5, max_
length=20)
summarized_40= summarizer(my_text, min_length=5, max_
length=40)
print(summarized_10)
print(summarized_20)
print(summarized_40)

The output is:
[{'summary_text': ' India won the World Cup on home'}]
[{'summary_text': ' India beat Sri Lanka in the 2011 World
Cup final at the Wankhede Stadium'}]
[{'summary_text': " India beat Sri Lanka in the 2011 World
Cup final at the Wankhede Stadium in Mumbai . MS Dhoni's
iconic six and Gautam Gambhir's brilliant knock were"}]
```

Translation from English to French: Let's try and do a bit of translation, from English to French.

```
en_to_fr = pipeline("translation_en_to_fr", model="Helsinki-
NLP/opus-mt-en-fr")
en_to_fr("Do you like reading story books?")

The output is:
[{'translation_text': "Vous aimez lire des livres d'histoires
?"}]
```

NLP will become even more popular in the coming years as a result of ready-to-use pretrained models and low-code, no-code technologies that are available to everyone. Businesses, in particular, will continue to gain from NLP, which will enable them to improve their operations, cut expenses, make smarter choices, and enable better customer experiences. Almost every communication we have with machines involves human communication, both spoken and written. NLP takes advantage of a deeper contextual understanding of the human language. END

References

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